

SPI Controller Verification Test Plan

A) APB/Register Correctness + Reset + Reserved Space

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-REG-01	R2 Reset values: all regs return specified reset after PRESETn	Directed: apply PRESETn low ≥ 2 cycles, release; read CTRL/STATUS/CLK_DIV/SS_CTRL/INT_EN/INT_STAT/DELAY	Coverpoint: reset observed per register	Scoreboard/direct check: APB reads compare vs reset constants
TP-REG-02	R1 R/W registers read back last written value (RO masked)	Directed + random: for each RW reg, write multiple patterns; read back	Per-register write/read bins; CTRL fields bins	Scoreboard compare on readbacks with RO mask
TP-REG-03	R22 APB always zero-wait: PREADY=1 always, PSLVERR=0 always	Always-on during all tests	Coverage: count of APB reads/writes; at least one access per reg	SVA bound to wrapper: PREADY==1, PSLVERR==0, plus APB protocol assertions
TP-REG-04	R23 Reserved offsets (0x24+) read 0, writes ignored	Directed: read several addresses $\geq 0x24$; write random then read back	Bin: reserved_read, reserved_write	Direct check: read data==0; no side-effects
TP-REG-05	R3 EN=0 holds shifter+FIFOs in reset; SCLK idle; SS forced high regardless SS_CTRL	Directed: set SS_CTRL low, set EN=0, write TX_DATA, observe SS_n/SCLK/BUSY	Bin: EN0 behavior exercised	SVA + scoreboard: SS high, SCLK idle, TX ignored

B) SPI Mode Correctness + Bit Order + Width Behavior + BUSY Timing

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-SPI-01	R4 SCLK idle polarity matches CPOL whenever BUSY=0	Directed sweep: MODE=0..3 idle and between transfers	Bin: 4 modes idle check	SVA: when BUSY==0, SCLK==CPOL
TP-SPI-02	R5 MOSI stable around sample edge; changes on launch edge per CPHA	Directed: each MODE, send known pattern	Bin: mode x edge type	SVA + monitor checks MOSI changes only on launch edges
TP-SPI-03	R6 MSB-first shifts [WIDTH-1] first; LSB-first shifts [0] first	Directed: patterns 0x81, 0x3C, 16'h8001, 32'h8000_0001 for both orders	Cross bins: mode $\times$ width $\times$ order	SPI monitor reconstructs stream; scoreboard compare
TP-SPI-04	R7 Transfer lasts exactly WIDTH SCLK cycles; BUSY asserted throughout; deasserts 1 PCLK after last sample edge	Directed: run 8/16/32, count edges	Width bins 8/16/32	Checker counts cycles; SVA optional
TP-SPI-05	R25 DIV/MODE/WIDTH/LSB_FIRST sampled at transfer start and held	Directed: start transfer, write fields during BUSY	Bin: mid-transfer write attempted per field	Current transfer unchanged; next transfer updated

C) Clock Divider Behavior

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-CLK-01	R8 SCLK = PCLK/(2*(DIV+1)) for all DIV [0..65535]	Directed corners + random DIV values	Coverage bins: corners + random	Count PCLK cycles per SCLK toggle
TP-CLK-02	R24 DIV=0 yields SCLK=PCLK/2	Directed: DIV=0 transfer	Bin: DIV==0 hit	Timing monitor check

D) FIFO Correctness

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-FIFO-01	R9 TX_DATA accepted while !TX_FULL; FIFO order preserved	Push N words (1,4,8) then transmit	TX occupancy bins	Scoreboard TX queue order
TP-FIFO-02	R11 TX depth exactly 8; TX_FULL asserts on 8th entry	Push 8 entries	Bin: TX full transition	STATUS read + scoreboard
TP-FIFO-03	R10 RX_DATA reads pop in FIFO order when !RX_EMPTY	Generate transfers then read RX_DATA repeatedly	RX occupancy bins	Scoreboard RX queue
TP-FIFO-04	R12 RX depth exactly 8; RX_FULL asserts on 8th received	8 transfers without reading	Bin: RX full transition	STATUS read + scoreboard
TP-FIFO-05	R13 TX write while full discarded, sets TX_OVF and INT_STAT	Fill TX then extra write	Bin: TX overflow event	Check discard + sticky flag
TP-FIFO-06	R14 Transfer while RX_FULL discards received word and sets RX_OVF	Fill RX then another transfer	Bin: RX overflow event	RX queue unchanged
TP-FIFO-07	R15 RX_DATA read while empty returns 0 and no RX_OVF	Read RX_DATA empty	Bin: RX empty read	Read==0; flags unchanged

E) SS_n Behavior

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-SS-01	R20 SS_n[i] = !SS_EN[i] OR SS_VAL[i] combinational	Toggle SS_CTRL while idle	Bin: each SS bit toggled	Direct compare SS_n equation
TP-SS-02	SS must remain asserted across transfer	Assert SS, start transfer, deassert mid-transfer	Bin: deassert attempt	Negative/undefined behavior monitor

F) Interrupt Controller Correctness

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-IRQ-01	R16 IRQ = OR(INT_STAT & INT_EN)	Cause each source masked/unmasked	Bin: each source masked/unmasked	SVA equivalence
TP-IRQ-02	R17 INT_STAT W1C behavior	Set bits then clear by writing 1	Bin: each bit set+cleared	Register reads
TP-IRQ-03	R18 W1C race: event coincident with clear => remains 1	Schedule clear same cycle as event	Bin: race case	Event wins
TP-IRQ-04	Interrupt sources coverage	Cause all five sources	Bin: all sources observed	Reads + scoreboard

G) Loopback Mode

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-LB-01	R19 LOOPBACK routes MOSI internally to RX; external MISO ignored	Enable LOOPBACK; random MISO; send words	Bin: width 8/16/32	RX==TX

H) Inter-Transfer Delay Behavior

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
TP-DLY-01	R21 DELAY idle half-cycles inserted between transfers while BUSY stays 1	Queue 2+ TX words; DELAY=0,1,>=128	Bin: DELAY values	Count idle half-cycles
TP-DLY-02	DELAY written during BUSY takes effect next transfer	Write DELAY mid-transfer	Bin: updated mid-busy	Current old, next new

Extra Scenarios

Label	Extra Scenario	Stimulus	Coverage	Check
X-01	Back-to-back transfers with SS held low	Queue many TX words	Long burst bin	SS never rises
X-02	Switch SS lines between transfers	Use SS0..SS3	Each SS used	SS mapping
X-03	INT_STAT precedence: multiple interrupts same time	Cause RX_FULL + DONE	Multi-bit set	Both bits set
X-04	TX_EMPTY interrupt edge correctness	Repeated empty transitions	Repeated transitions	Expected sticky behavior
X-05	Random DIV + DELAY + mode + width	Constrained random loops	Random bins	Scoreboard + SVA
X-06	RX_DATA reads during BUSY	Read during active transfer	Read-during-busy bin	Pops only on read
X-07	CTRL.EN 1→0 flushes FIFOs but not INT_STAT	Pending TX/RX then disable	en-toggle flush	FIFOs empty, INT_STAT kept
X-08	Loopback + LSB-first + width=32	Combine corners	Corner bin	RX==TX